

C++ in Quantitative Finance

#10

C++ in Excel: Pricing path dependent options

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Exercises I

1. Implement in Excel a pricer for the arithmetic/geometric Asian options. Your function should have another argument to decide about the type of Asian option. Propose some plots representing how option price relates to function arguments.
2. Add to your pricer a new argument - whether to use or not antithetic variates.
3. How would you value a typical European Call option using your option pricer?

Exercises II

4. Write and implement a pricer for barrier options. Here are some hints:

- Take the `cpp09` solution as the starting point.
- Design a new class: `PathDependentBarrier` by copying and modifying the existing `PathDependentAsian` class.
- Add to the `PathDependentBarrier` class two new data members: `double Barrier` and `short int Flavour`. The flavour should be defined as: 1 for up-and-in, 2 for up-and-down, 3 for down-and-in and 4 for down-and-out.
- Modify the constructor of the `PathDependentBarrier` class.
- Modify the `PathDependentBarrier::CashFlows(const MJArray&SpotValues, std::vector<CashFlow>&GeneratedFlows)` method.
- In the `main()` function, ask the user to provide the barrier level and the flavour of the barrier option.
- Create an object of the `PathDependentBarrier` class and put it to the `theEngine`, using the same way as previously for `PathDependentAsian` class.

Thank you!

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